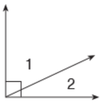
Angle Relationships

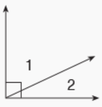
The right angle shown here is divided into two angles, ∠1 and ∠2. The sum of the measures of the two angles must be 90°.



Two angles that add up to 90° are called **complementary angles.** If you know the measure of one angle, you can find the measure of the other.

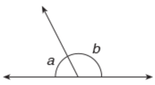
Example 1

Hide



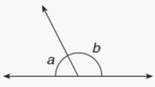
In the drawing, $m\angle 2 = 25°$. What is the measure of ∠1?
You know that $m\angle 1 + m\angle 2 = 90°$. Therefore, $m\angle 1 + 25° = 90°$.
Subtract to find the measure of ∠1: $90° - 25° = 65°$
$m\angle 1 = 65°$

The straight angle shown here is divided into two angles, ∠*a* and ∠*b*. The sum of the measures of the angles must be 180°. Two angles that add up to 180° are called **supplementary angles.**



Example 2

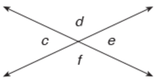
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If ∠*a* measures 63°, what is the measure of ∠*b*?
Subtract to find the measure of ∠*b*: $180° - 63° = 117°$
$m\angle b = 117°$

When straight lines intersect, they form pairs of supplementary angles. Look at the drawing. How many pairs of supplementary angles do you see?

There are four pairs: ∠*c* and ∠*d*, ∠*d* and ∠*e*, ∠*e* and ∠*f*, ∠*f* and ∠*c*.

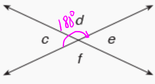


Each pair of angles can also be described as adjacent. **Adjacent angles** share a side. Angles that do not share a side but are opposite each other are **vertical angles.** Vertical angles have equal measures. also called opposite angles.

When two lines intersect in this way, you need the measure of only one of the angles to find the measures of the others.

Example 3

Hide



If the measure of ∠*c* is 50°, what is the measure of each of the other angles?
Since ∠*c* and ∠*d* are supplementary, the measure of ∠*d* is $180° - 50° =$ **130°**.
Since ∠*c* and ∠*e* are vertical angles, the measure of ∠*e* is also **50°**.
Since ∠*d* and ∠*f* are vertical angles, the measure of ∠*f* is also **130°**.

In geometry you must have a reason for everything you do. You can't conclude that two angles are equal just because they look equal. You have to have evidence that they are equal. Good test takers learn the definitions of geometry terms, because the definitions can be used as evidence that something is true.

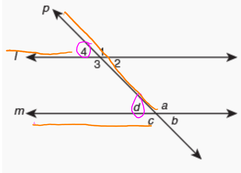
When one line crosses another line, vertical angles are formed. When one line crosses two parallel lines, the line, called a **transversal,** creates vertical angles in two places. These are called **corresponding angles** because they are in the same position on the parallel lines. Corresponding angles are equal. If you know one angle, you can find the others.

Example 4

Hide

∠d = ∠4 (F-pattern)

TPT - transverse parallel theorem



The measure of ∠4 is 40°. Find all the other angles.

Conclusion	Reason	Conclusion	Reason
∠2 = 40°	∠4 and ∠2 are vertical angles.	∠*d* = 40°	∠4 and ∠*d* are corresponding angles.
∠1 = 140°	∠1 and ∠2 are supplementary.	∠*b* = 40°	∠*b* and ∠*d* are vertical angles.
∠3 = 140°	∠1 and ∠3 are vertical.	∠*a* = 140°	∠1 and ∠*a* are corresponding angles.
		∠*c* = 140°	∠*a* and ∠*c* are vertical angles.

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